Confusion Is Not Frustration: Distinguishing Epistemic Affect in Chat-Based Tutoring Logs

Marco Gallo, Hedda Stein

In Proceedings of the 16th International Conference on Educational Data Mining (EDM '23), pp. 77-88.

Abstract

Detectors of negative affect in tutoring logs often merge confusion and frustration. We annotated 6,200 learner turns from a chat-based statistics tutor with separate labels and found that the two states have different textual signatures and different consequences: confusion followed by a resolving explanation predicted higher post-test scores, while frustration did not. A detector that separates the two improves the usefulness of affect-triggered interventions.

1 Introduction

Confusion arises when new information conflicts with what a learner expects; it can be productive if resolved. Frustration arises when goals are blocked; it tends to persist. Treating both as "negative affect" loses this difference.

2 Annotation

Three annotators labeled 6,200 learner turns as confused, frustrated, both, or neither, with access to the preceding three turns. Agreement was kappa 0.61 for confusion and 0.57 for frustration.

Label	Turns	Typical cues
Confused	1,140	questions with "but", hedges, restating the tutor's words
Frustrated	610	repetition, giving up, complaints about the tutor
Both	230	confused question followed by an exasperated aside
Neither	4,220	-

3 Consequences

Confusion that was followed within two turns by an explanation the learner acknowledged predicted a 6-point higher post-test score compared with no confusion. Unresolved confusion predicted no difference. Frustration predicted lower scores whether or not the tutor responded.

4 Detector

A two-headed classifier reached F1 0.71 (confusion) and 0.63 (frustration). A merged negative-affect detector had higher overall F1 (0.74) but triggered interventions on confused learners who did not need them.

5 Implications

Respond to confusion with explanation and to frustration with a change of task or pace. Do not rush to remove confusion.

References

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[2] Baker, R. S. J. d. et al. Better to be frustrated than bored. International Journal of Human-Computer Studies 68, 4 (2010), 223-241.